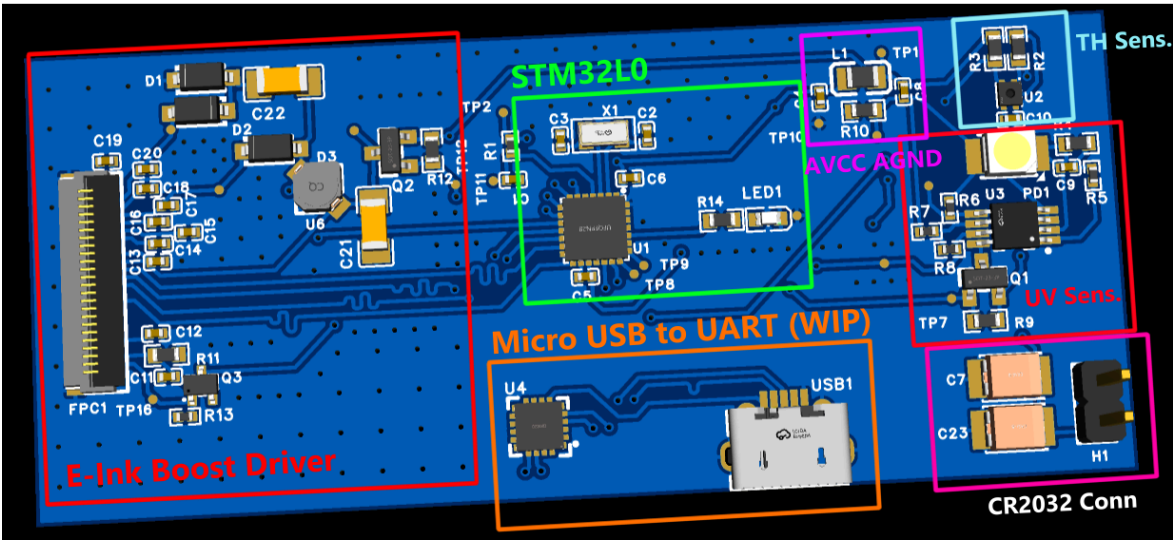
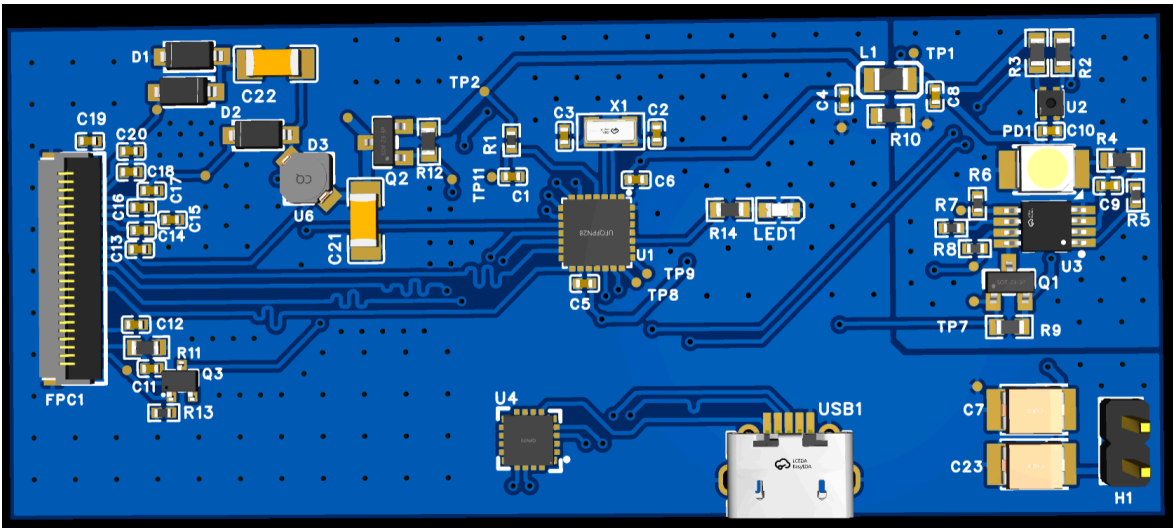
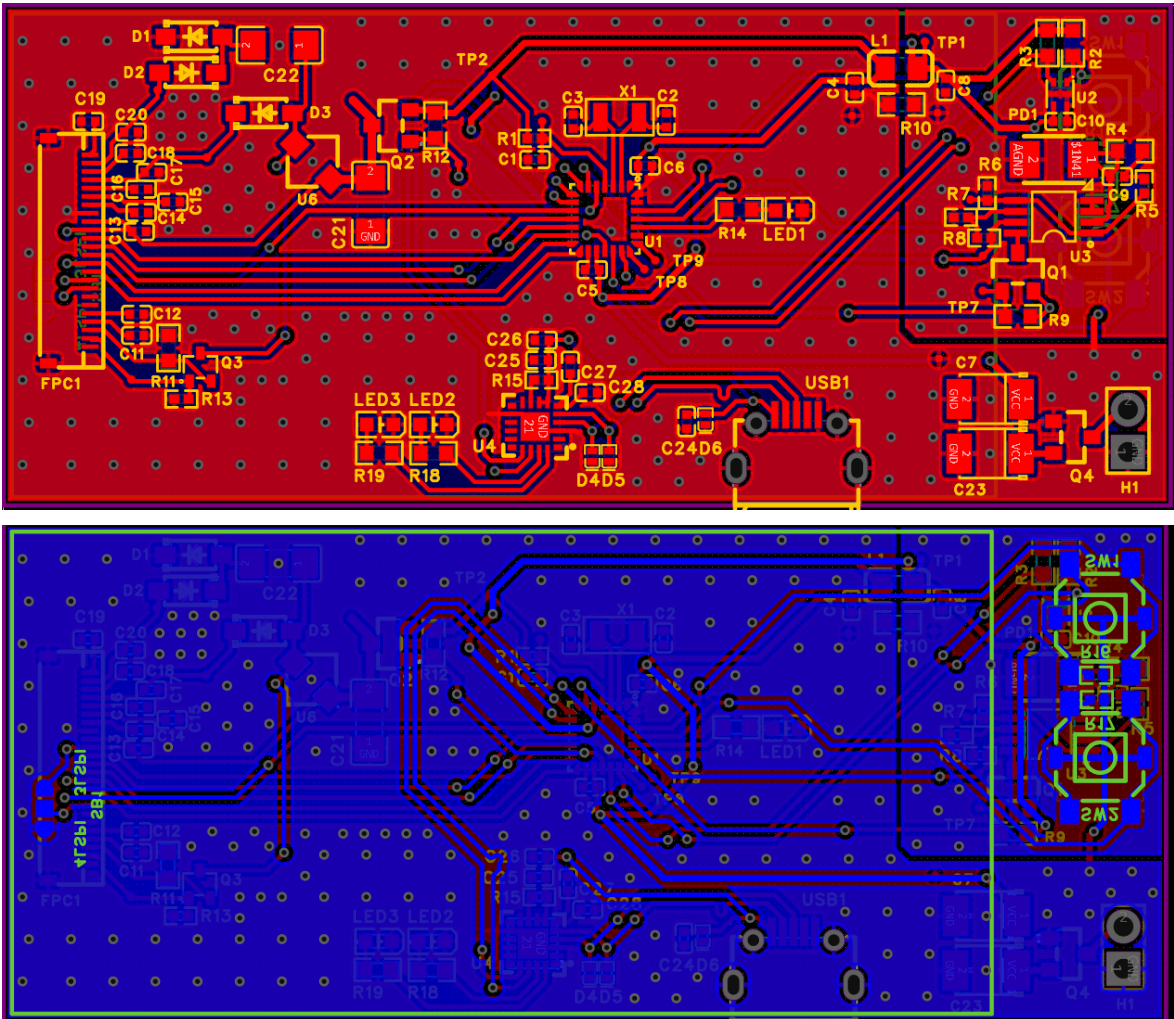


AtmosTag





*note: pictures shown above are for illustrative purpose and don't represent final design

The AtmosTag is an ultra-low-power, pocket-sized device featuring a 2.13-inch e-ink display, designed for persistent visual data presentation and environmental monitoring on a single coin cell battery. Built around a zero-revision engineering philosophy, the project merges low-level embedded firmware with a streamlined, cost-effective single-board layout.

Key Technical Specifications

- Microcontroller:** STM32L031 (ARM Cortex-M0+) running LL API for maximum execution speed and fine-grained sleep-state control.
- Display:** 2.13-inch monochrome e-ink panel driven by an on-board SSD1680 controller circuit.
- Environmental Sensing:** SHT41 high-precision temperature and humidity sensor. GUA-S12SD with custom front-end signal conditioning to monitor UV Index.
- Power & Boost Architecture:** Powered by a CR2032 coin cell battery, featuring a tightly clustered high-voltage boost converter (SSD1680 reference design generating +22V VGH and -20V VGL rails) with 25v–50v rated output capacitors to prevent dielectric breakdown.
- Connectivity & Updates:** micro usb connector paired with a CP2102 USB-to-UART bridge, allowing users to flash custom 4.7 KB bitmap images and update configurations via a lightweight desktop utility.

Main Components:

- STM32L031-G6U6
- SHT41
- GUVA-S12SD -> MCP6002
- CP2102

Timeline

Month	Week	Description
September	3-5	• Schematic Design • Proteus Simulations • PCB Layout Prototype
October	1-2	• Enclosure Design • Finish PCB Layout Design
October	3-4	• Design Verification • PCBA Cost Planning • Finish Enclosure Design
November	1	• Final Checks
November	2-4	• Manufacturing + Shipment • Firmware Prototyping on Nucleo
December	1-2	• PCB Touchup + Testing • Firmware Testing
December	3-4	• Finish Firmware • Develop Software
January	1-4	Polish software & firmware

Constraints

- Maximum Budget: IDR XXX (includes manufacturing costs, shipments, taxes, tools, etc.)
- Maximum Deadline: By the end of January
- First-Time-Right PCB:
 - PCB Shall not be redesigned for it to work after being manufactured
 - Reworks (replace components, jumpers, etc.) are acceptable
- The whole circuitry is powered by 200 mAh Coin Cell Battery (CR2032) and have minimum life expectancy of 9 months
- The whole circuitry, screen, and its enclosure should fit into a pocket (max: 4cm * 10cm * 2cm)

Project Funding

Event name : EasyEDA Sparks 2026

Description :

- JLC Group will fund 80% of material and shipping costs
- The design must be open sourced and made using EasyEda PRO
- Fund will be granted after review by their team
- Cost estimation:
 - 50-70 USD before (PCBA) funding
 - 12-13 USD after (PCBA) funding

All (3)JLCPCB (3)JLC3DP (0)JLCCNC (0)JLCMC (0)Flex Heater (0)

Search

☐	Item	Qty	Build Time	Price	
☒	JLCPCB (PCB/PCBA/Stencil)				
☒	 AtmosTag_AtmosTag-v1.0_20260920_... PCB prototype:Y9-13464662A Green, 1.6 Thickness, HASL(with lead) Customer #:  Product Details Edit Order	5	2 days	\$4.00	
	 AtmosTag_AtmosTag-v1.0_20260920_... Economic PCBA: SMT026100460946-1... Assemble top side Customer #:  Product Details	2	1 - 2 days	\$52.47	

SUMMARY

[2 items>](#)

Merchandise Total

\$56.47

Shipping Estimated

\$9.88

Coupons

-\$10.00

Subtotal

\$56.35

Est. shipping date

2026-10-07

Order won't be shipped until all items are ready.

Weight

0.26kg

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